EE133 HW 8

MOS Capacitor, CV Characteristics, MOSFET Short-Channel Effects

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Problem 1

Given: Ideal MOS capacitor, $t_{ox} = 5\,\text{nm}$, $\varepsilon_{ox} = 3.9\,\varepsilon_0$, p-type Si with $N_A = 5 \times 10^{16}\,\text{cm}^{-3}$, $\chi_{Si} = 4.05\,\text{eV}$, $E_g = 1.12\,\text{eV}$.
Constants used throughout:

$$\begin{aligned}
\varepsilon_0 &= 8.854 \times 10^{-14}\,\text{F/cm} \\
\varepsilon_{Si} &= 11.7\,\varepsilon_0 = 1.036 \times 10^{-12}\,\text{F/cm} \\
\varepsilon_{ox} &= 3.9\,\varepsilon_0 = 3.453 \times 10^{-13}\,\text{F/cm} \\
k_BT &= 0.02585\,\text{eV} \quad (T = 300\,\text{K}) \\
n_i &= 1.5 \times 10^{10}\,\text{cm}^{-3}
\end{aligned}$$

(a) Ideal MOS Capacitor

Step 1: Fermi potential ϕ_F

$$\phi_F = \frac{k_BT}{q}\ln\left(\frac{N_A}{n_i}\right) = 0.02585\ln\left(\frac{5 \times 10^{16}}{1.5 \times 10^{10}}\right) = 0.02585\ln\left(3.333 \times 10^6\right) \approx \boxed{0.3945\,\text{V}} \tag{1}$$

(i) Maximum depletion width $W_{d,\text{max}}$:
Strong inversion occurs when $\phi_s = 2\phi_F$:

$$\begin{aligned}
W_{d,\text{max}} &= \sqrt{\frac{2\,\varepsilon_{Si}\,(2\phi_F)}{q\,N_A}} = \sqrt{\frac{2 \times (1.036 \times 10^{-12}) \times (2 \times 0.3945)}{(1.602 \times 10^{-19})(5 \times 10^{16})}} \\
&= \sqrt{\frac{2 \times 1.036 \times 10^{-12} \times 0.789}{8.01 \times 10^{-3}}} = \sqrt{\frac{1.634 \times 10^{-12}}{8.01 \times 10^{-3}}} = \sqrt{2.040 \times 10^{-10}} \\
&\approx \boxed{1.43 \times 10^{-5}\,\text{cm} = 143\,\text{nm}}
\end{aligned} \tag{2}$$

(ii) Oxide and minimum capacitances:

$$C_{\text{ox}} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.453 \times 10^{-13}}{5 \times 10^{-7}} = 6.906 \times 10^{-7}\,\text{F/cm}^2 \tag{3}$$

$$C_{dep,\text{min}} = \frac{\varepsilon_{Si}}{W_{d,\text{max}}} = \frac{1.036 \times 10^{-12}}{1.43 \times 10^{-5}} = 7.245 \times 10^{-8}\,\text{F/cm}^2 \tag{4}$$

$$\begin{aligned}
C_{\text{min}} &= \frac{C_{\text{ox}} \cdot C_{dep,\text{min}}}{C_{\text{ox}} + C_{dep,\text{min}}} = \frac{6.906 \times 10^{-7} \times 7.245 \times 10^{-8}}{6.906 \times 10^{-7} + 7.245 \times 10^{-8}} \\
&\approx \boxed{6.26 \times 10^{-8}\,\text{F/cm}^2}
\end{aligned} \tag{5}$$

(iii) Threshold voltage (ideal, $V_{FB} = 0$):
Maximum depletion charge:

$$Q_{d,\text{max}} = -q\, N_A\, W_{d,\text{max}} = -(1.602 \times 10^{-19})(5 \times 10^{16})(1.43 \times 10^{-5}) = -1.145 \times 10^{-7}\ \text{C/cm}^2 \quad (6)$$

$$\begin{aligned} V_t &= V_{FB} - \frac{Q_{d,\text{max}}}{C_{\text{ox}}} + 2\phi_F = 0 + \frac{1.145 \times 10^{-7}}{6.906 \times 10^{-7}} + 2(0.3945) \\ &= 0.1658 + 0.789 \approx \boxed{0.955\,\text{V}} \end{aligned} \quad (7)$$

(b) p$^+$ Polysilicon Gate with Fixed Surface Charge

Determine ϕ_{ms}:
For p^+ polysilicon, E_F is at the valence band edge, so:

$$\phi_m(\text{p}^+\ \text{poly}) = \chi_{Si} + E_g/q = 4.05 + 1.12 = 5.17\,\text{eV (below vacuum)} \quad (8)$$

For p-type Si with $N_A = 5 \times 10^{16}$:

$$\phi_s = \chi_{Si} + \frac{E_g}{2q} + \phi_F = 4.05 + 0.56 + 0.3945 = 5.005\,\text{eV} \quad (9)$$

$$\phi_{ms} = \phi_m - \phi_s = 5.17 - 5.005 = \boxed{+0.165\,\text{V}} \quad (10)$$

(i) Flat band voltage V_{FB}:
Fixed surface charge $Q_{ss} = (1.602 \times 10^{-19})(10^{11}) = 1.602 \times 10^{-8}$ C/cm^2 (positive).

$$\begin{aligned} V_{FB} &= \phi_{ms} - \frac{Q_{ss}}{C_{\text{ox}}} = 0.165 - \frac{1.602 \times 10^{-8}}{6.906 \times 10^{-7}} \\ &= 0.165 - 0.02320 \approx \boxed{0.142\,\text{V}} \end{aligned} \quad (11)$$

(ii) Threshold voltage with V_{FB}:

$$V_t = V_{FB} - \frac{Q_{d,\text{max}}}{C_{\text{ox}}} + 2\phi_F = 0.142 + 0.1658 + 0.789 \approx \boxed{1.097\,\text{V}} \quad (12)$$

(c) Al Gate

From slide 20 data: $\phi_m(\text{Al}) \approx 4.1$ eV, $\phi_s = 5.005$ eV.

$$\phi_{ms}(\text{Al}) = 4.1 - 5.005 = \boxed{-0.905\,\text{V}} \quad (13)$$

$$V_t(\text{Al}) = \phi_{ms} + \left(-\frac{Q_{d,\text{max}}}{C_{\text{ox}}}\right) + 2\phi_F = -0.905 + 0.1658 + 0.789 \approx \boxed{0.050\,\text{V}} \quad (14)$$

(d) Pt Gate

From slide 20 data: $\phi_m(\text{Pt}) \approx 5.65\,\text{eV}$.

$$\phi_{ms}(\text{Pt}) = 5.65 - 5.005 = \boxed{+0.645\,\text{V}} \tag{15}$$

$$V_t(\text{Pt}) = 0.645 + 0.1658 + 0.789 \approx \boxed{1.600\,\text{V}} \tag{16}$$

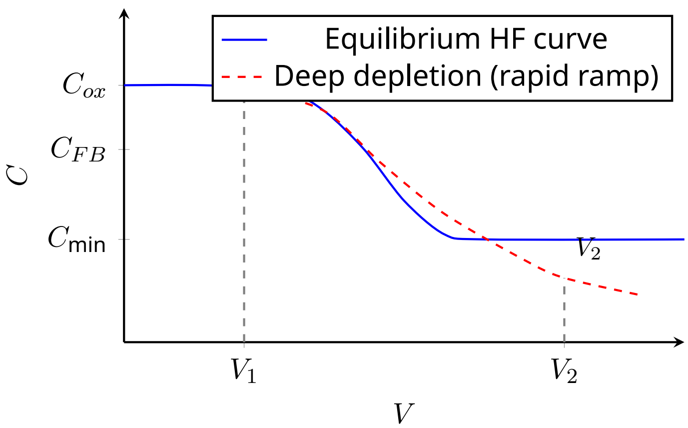
Problem 2 — Rapid Voltage Ramp from V_1 to V_2

Situation: The device is biased in strong inversion at V_1 (left of threshold on a p-type MOS), then rapidly ramped to deep depletion at V_2 (far right, beyond threshold).

Explanation:

At V_1 the device is in accumulation or weak inversion. The initial capacitance is high ($\approx C_{ox}$). When the gate voltage is *rapidly* ramped to V_2, the system is driven deep into depletion because the inversion layer **cannot form instantaneously** — minority carrier (electron) generation from the bulk is a slow thermal process with time constant $\tau \sim \mu$s–ms.

As a result, the depletion region continues to **widen beyond** $W_{d,\text{max}}$, following the *deep depletion* CV curve rather than the equilibrium high-frequency curve. The capacitance drops *below* C_{min} and continues to decrease as V is held at V_2.



The CV trace follows the **deep depletion** path (red dashed), where $C < C_{\text{min}}$, because minority carrier generation cannot keep up with the fast sweep rate. Over time (seconds to minutes), thermally generated minority carriers fill in the inversion layer and the capacitance recovers to C_{min}.

Problem 3 — CV Curve Analysis

From the given high-frequency CV plot: $C_{ox} = 20\,\text{pF}$, $C_{\text{min}} \approx 5\,\text{pF}$, $V_{FB} \approx +1\,\text{V}$ (indicated on plot), and the CV curve is shifted to the right (positive gate voltages required for inversion $\Rightarrow$ p-type substrate, n-channel device).

(a) ϕ_{ms} from Ideal Shift

The ideal CV curve for p-type Si (no charges, no workfunction difference) is centered near $V = 2\phi_F > 0$, peaking at $V_{FB} = 0$. Here $V_{FB} \approx +1\,\text{V}$.

If the shift is *only* due to ϕ_{ms}:

$$V_{FB} = \phi_{ms} \quad \text{(no charges)} \tag{17}$$

$$\therefore\ \phi_{ms} \approx \boxed{+1.0\,\text{V}} \tag{18}$$

(b) Oxide Thickness from C_{ox}

Gate area $A = 10^{-3}\,\text{cm}^2$, $C_{ox} = 20\,\text{pF} = 20 \times 10^{-12}\,\text{F}$.

$$C''_{ox} = \frac{C_{ox}}{A} = \frac{20 \times 10^{-12}}{10^{-3}} = 2 \times 10^{-8}\,\text{F/cm}^2 \tag{19}$$

$$t_{ox} = \frac{\varepsilon_{ox}}{C''_{ox}} = \frac{3.453 \times 10^{-13}}{2 \times 10^{-8}} = 1.727 \times 10^{-5}\,\text{cm} \approx \boxed{172.7\,\text{nm}} \tag{20}$$

(c) Maximum Depletion Width with $N_A = 10^{16}$ cm^{-3}

$$\phi_F = 0.02585\,\ln\left(\frac{10^{16}}{1.5 \times 10^{10}}\right) = 0.02585 \times 13.41 \approx 0.3466\,\text{V} \tag{21}$$

$$\begin{aligned} W_{d,\text{max}} &= \sqrt{\frac{2\,\varepsilon_{Si}\,(2\phi_F)}{q\,N_A}} = \sqrt{\frac{2 \times 1.036 \times 10^{-12} \times 0.6932}{1.602 \times 10^{-19} \times 10^{16}}} = \sqrt{\frac{1.436 \times 10^{-12}}{1.602 \times 10^{-3}}} \\ &= \sqrt{8.963 \times 10^{-10}} \approx \boxed{2.994 \times 10^{-5}\,\text{cm} = 299\,\text{nm}} \end{aligned} \tag{22}$$

(d) Effect of Increased Substrate Doping on V_{FB} and $W_{d,\text{max}}$

- V_{FB} shifts because ϕ_{ms} depends on $\phi_s = \chi_{Si} + E_g/2q + \phi_F$. Higher $N_A \Rightarrow$ larger ϕ_F $\Rightarrow$ larger $\phi_s \Rightarrow \phi_{ms}$ becomes more negative (for a fixed metal gate). V_{FB} will therefore **decrease** (shift more negative).
- $W_{d,\text{max}} \propto \sqrt{2\phi_F/N_A} \propto \sqrt{\ln(N_A)/N_A}$. Since N_A grows faster than $\ln(N_A)$, $W_{d,\text{max}}$ **decreases** with increasing doping.

(e) Threshold Voltage (approximate)

From the graph, the CV curve drops sharply and levels off at C_{min} around $V \approx 2\,\text{V}$. Reading directly from the plot:

$$V_t \approx \boxed{+2.0\,\text{V}} \tag{23}$$

(f) Implant to Reduce V_t by 1 V

We want $\Delta V_t = -1\,\text{V}$ by implanting a sheet charge Q_I at the Si/SiO_2 interface.

A sheet charge at the oxide/Si interface shifts V_t by $-Q_I/C''_{ox}$:

$$\Delta V_t = -\frac{q\,N_I}{C''_{ox}} = -1\,\text{V} \tag{24}$$

$$N_I = \frac{C''_{ox} \times 1\,\text{V}}{q} = \frac{2 \times 10^{-8}}{1.602 \times 10^{-19}} \approx \boxed{1.25 \times 10^{11}\ \text{cm}^{-2}} \tag{25}$$

To *decrease* V_t (which is positive for this p-type, n-channel device), we need a **positive** charge at the interface. A donor dopant (phosphorus, **P**) implanted at the interface provides positive fixed charge and reduces V_t.

Problem 4 — Effect of Decreasing Oxide Thickness

Device: n-channel MOSFET (p-type substrate), $Q_{ss} = 1.602 \times 10^{-9}$ C/cm^2.

(a) Flat-Band Voltage — Decrease

$$V_{FB} = \phi_{ms} - \frac{Q_{ss}}{C_{ox}} = \phi_{ms} - \frac{Q_{ss}\, t_{ox}}{\varepsilon_{ox}} \tag{26}$$

V_{FB} includes the term $-Q_{ss}/C_{ox} = -Q_{ss}\, t_{ox}/\varepsilon_{ox}$. Since $Q_{ss} > 0$, this term is negative. As t_{ox} **decreases**, C_{ox} **increases**, so $|Q_{ss}/C_{ox}|$ **decreases** (becomes less negative). Therefore V_{FB} **increases** toward ϕ_{ms}.

Answer: decrease $\rightarrow$ actually increase

Explanation: $V_{FB} = \phi_{ms} - Q_{ss}/C_{ox}$. Positive Q_{ss} makes $V_{FB} < \phi_{ms}$. Decreasing t_{ox} raises C_{ox}, making the Q_{ss}/C_{ox} correction term smaller in magnitude. Thus V_{FB} **increases** (becomes less negative / larger). **Circle: increase.**

(b) Threshold Voltage — Increase

$$V_t = V_{FB} + 2\phi_F - \frac{Q_{d,\text{max}}}{C_{ox}} \tag{27}$$

With decreasing t_{ox}: $C_{ox} \uparrow$, so $Q_{d,\text{max}}/C_{ox} \downarrow$, and V_{FB} increases (as shown above). Both effects raise V_t. More precisely, the dominant term $-Q_{d,\text{max}}/C_{ox}$ (negative for p-type substrate in n-channel device) becomes less negative $\Rightarrow V_t$ **increases**.

Circle: increase.

Problem 5 — Threshold Voltage Shift from Oxide Charges

Start from the ideal MOS capacitor: $V_t = 2\phi_F - Q_{d,\text{max}}/C_{ox}$.

(a) Uniform Volume Charge ρ_{ox} in Oxide

Consider a thin slab of charge $dQ = \rho_{ox}\,dx$ at position x from the metal gate (oxide spans $0 \le x \le t_{ox}$). Its contribution to the voltage shift at the gate is:

$$d(\Delta V_t) = -\frac{\rho_{ox}\,dx}{\varepsilon_{ox}} \cdot \frac{t_{ox} - x}{t_{ox}} \tag{28}$$

(The factor $(t_{ox} - x)/t_{ox}$ accounts for the fraction of the oxide field that appears at the gate.)
Integrating over the full oxide:

$$\Delta V_t = -\frac{\rho_{ox}}{\varepsilon_{ox}} \int_0^{t_{ox}} \frac{t_{ox} - x}{t_{ox}}\,dx = -\frac{\rho_{ox}}{\varepsilon_{ox}} \cdot \frac{t_{ox}}{2}$$

$$\boxed{\Delta V_t = -\frac{\rho_{ox}\,t_{ox}}{2\,\varepsilon_{ox}} = -\frac{\rho_{ox}\,t_{ox}}{2\,C_{ox}\,t_{ox}} = -\frac{\rho_{ox}}{2\,C_{ox}}} \tag{29}$$

A positive uniform oxide charge ($\rho_{ox} > 0$) shifts V_t **negative**.

(b) Sheet Charge Q_s at Metal–SiO$_2$ Interface ($x = 0$)

A sheet charge at the metal interface ($x = 0$) is the farthest from the Si surface. Its weighting factor $(t_{ox} - 0)/t_{ox} = 1$, so it contributes maximally to the gate voltage but *zero* effect on the semiconductor because the entire potential drop is across the oxide:

$$\Delta V_t = -\frac{Q_s}{\varepsilon_{ox}} \cdot t_{ox} \cdot \frac{1}{t_{ox}} = -\frac{Q_s}{\varepsilon_{ox}/t_{ox}} \tag{30}$$

$$\boxed{\Delta V_t = -\frac{Q_s}{C_{ox}}} \tag{31}$$

Wait — more precisely: a sheet at $x = 0$ (metal side) shifts the flat-band voltage by the full $-Q_s/C_{ox}$, which shifts V_t by the same amount. A sheet at $x = t_{ox}$ (Si side) shifts V_t by $-Q_s/C_{ox}$ as well but is the standard interface charge case.

Summary:

Charge location	ΔV_t	Note
Uniform in oxide ρ_{ox}	$-\dfrac{\rho_{ox}\,t_{ox}}{2\varepsilon_{ox}}$	Half-weight average
Sheet at metal ($x = 0$)	$-\dfrac{Q_s}{C_{ox}}$	Full shift

Problem 6 — Subthreshold Slope Check

Claim: I_D changes by 10^4 over $\Delta V_G = 200\,\text{mV}$.

The subthreshold swing S describes how many mV of V_G are needed to change I_D by one decade:

$$S = \frac{dV_G}{d(\log_{10} I_D)} = \frac{kT}{q}\ln(10)\left(1+\frac{C_{dep}}{C_{ox}}\right) \geq \frac{kT}{q}\ln(10) \approx 60\,\text{mV/decade} \quad (T = 300\,\text{K}) \tag{32}$$

The theoretical **minimum** subthreshold swing at room temperature is 60 mV/decade (when $C_{dep}/C_{ox} \to 0$).

If I_D changes by 10^4 (i.e., 4 decades) over 200 mV:

$$S_{\text{claimed}} = \frac{200\,\text{mV}}{4\,\text{decades}} = 50\,\text{mV/decade} \tag{33}$$

Since 50 mV/decade < 60 mV/decade, this **violates the thermal limit**.

Tell him to go measure again.

The minimum subthreshold swing at 300 K is ≈ 60 mV/decade regardless of device geometry. A slope of 50 mV/decade is physically impossible in a conventional MOSFET at room temperature. The measurement likely has a calibration error or noise floor issue.

Problem 7 — DIBL (Drain-Induced Barrier Lowering)

DIBL (Drain-Induced Barrier Lowering) is a short-channel effect where the electric field from the drain depletion region penetrates into the channel region and lowers the source-to-channel potential barrier.

Primary effects:

1. **Threshold voltage reduction:** V_t decreases as V_{DS} increases. In long-channel devices, V_{DS} has negligible effect on V_t; in short-channel devices, a high V_{DS} can significantly lower V_t, parametrized as $\Delta V_t = -\text{DIBL} \times V_{DS}$ where DIBL is in mV/V.

2. **Increased off-state leakage (I_{off}):** Because V_t is lowered at high V_{DS}, the subthreshold current increases even when $V_{GS} = 0$, degrading the off-state isolation of the transistor.

3. **Reduced gate control:** The gate loses electrostatic control over the channel potential because the drain field competes with the gate field. This degrades the subthreshold slope and leads to a "punchthrough" condition at extreme cases where the drain field fully depletes the channel regardless of gate bias.

4. **Increased I_{on}/I_{off} ratio degradation:** Since I_{off} increases and device behavior becomes V_{DS}-dependent, the usable logic swing and noise margin shrink.

DIBL is quantified as: $\text{DIBL} = -\frac{\Delta V_t}{\Delta V_{DS}}$ in mV/V. Values above ~ 100 mV/V indicate serious short-channel behavior.

Problem 8 — GIDL (Gate-Induced Drain Leakage)

GIDL (Gate-Induced Drain Leakage) occurs when a large negative gate-to-drain voltage ($V_{GD} \ll 0$ for n-channel) creates a large electric field in the gate-oxide–drain overlap region, inducing **band-to-band tunneling (BTBT)** in the drain.

Primary effects:

1. **Off-state leakage current:** When $V_G = 0$ (off) and $V_D > 0$, the overlap region sees $V_{GD} < 0$. This bends the drain bands steeply, allowing electrons to tunnel from the valence band to the conduction band (or equivalently, holes to tunnel from conduction to valence band). The resulting hole current flows into the bulk and electron current into the drain, creating a leakage path even when the channel is off.

2. **Increased standby power:** Since GIDL is an off-state leakage mechanism, it directly increases static power dissipation in CMOS circuits, a critical concern for battery-powered devices.

3. **Body potential modulation (floating body):** In SOI or floating-body devices, GIDL injects holes into the floating body, raising its potential. This can lower V_t (forward-biasing body-source junction) and further increase leakage in a positive feedback loop.

4. **Worse at thin oxides and high V_{DD}:** GIDL scales with the electric field in the overlap region, so reducing t_{ox} or increasing supply voltage worsens GIDL. Unlike DIBL, GIDL is particularly sensitive to V_{DG}, not V_{DS} alone.

GIDL is distinct from DIBL in that it is a quantum-mechanical tunneling process localized at the drain-gate overlap, not a classical electrostatic barrier-lowering effect.

Problem 9 — Long-Channel or Short-Channel Device?

The I-V characteristic is from a long-channel device.
Evidence and reasoning:

1. **Clear saturation:** The drain current I_D saturates and becomes nearly flat for $V_D > V_{DS,\text{sat}} = V_{GS} - V_t$. In long-channel devices, I_D in saturation follows $I_D \propto (V_{GS} - V_t)^2$ (square-law), independent of V_{DS}. Short-channel devices exhibit velocity saturation which causes I_D to saturate earlier and at lower V_{DS}.

2. **Quadratic I_{DS}–V_{GS} dependence:** The equal spacing between curves at $V_G = -0.5, -1.0, -1.5$ V (current roughly quadruples from -0.5 V to -1.0 V and then doubles again to -1.5 V relative to threshold) is consistent with a square-law $(V_{GS} - V_t)^2$ relationship. Short-channel devices show a more *linear* (first-power) dependence due to velocity saturation.

3. **No channel-length modulation or DIBL:** The saturation curves are flat with no visible upward slope in saturation. A short-channel device would show output conductance ($\partial I_D / \partial V_D \neq 0$ in saturation) due to DIBL and channel-length modulation.

4. **Negative V_G curves for conduction** suggests a p-channel device, consistent with a conventional long-channel PMOS where the square-law model holds well.

Conclusion: The sharply defined saturation, quadratic V_{GS} dependence, and flat saturation region are hallmarks of the long-channel gradual-channel approximation model. A short-channel device would show velocity saturation (linear I_{DS}–V_{GS}), DIBL, and non-flat saturation curves.